COMP 3700A: Contributing to Open Source

~~Exploring OSS projects~~

Git Concepts

Charlotte Curtis

September 14, 2026

Today's Goals

- Understand a bit more about Git

  ~~- Look at the projects you shared for homework:

  - Common documents
  - Project activity
  - Communication channels~~

References:

- "Open Source: What It Is and How to Contribute" — Chapter 4 (Git)

Housekeeping

- Lab grades will be updated on D2L out of seven
- Lab deadlines now made explicit on labs README, with extensions for first two
- Course notes are now live
- Project part 1 is now published

> *I asked Gemini and it was completely willing to invent answers. Please don't waste my time (and yours) outsourcing this to an LLM.*

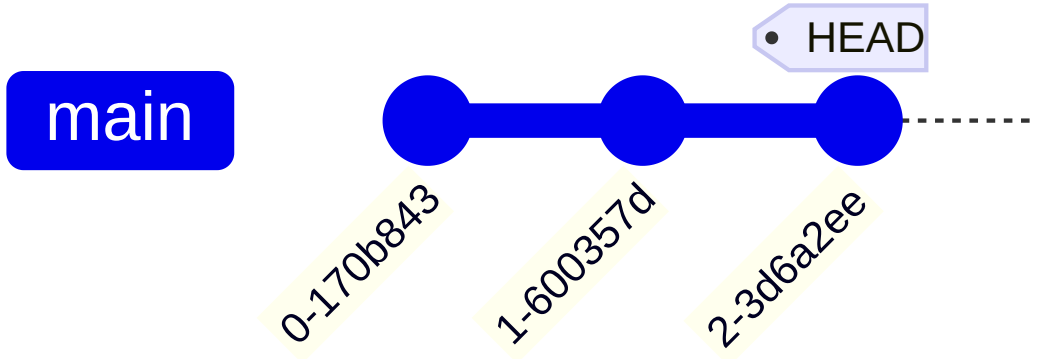
Single user, single branch

- `init` or `clone` to get repo
- For each commit:

```
git add whatever changed
git commit -m "Short message \
  describing change"
```

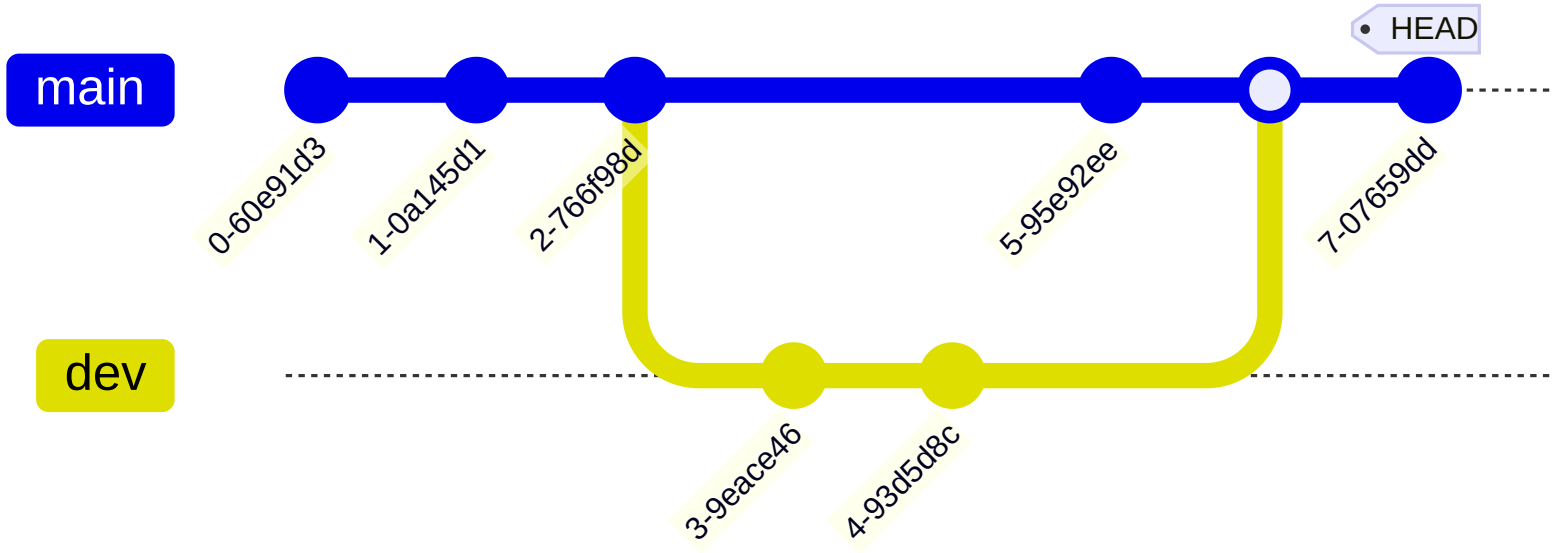
- Other useful commands:

```
git status
git log
git reset
```

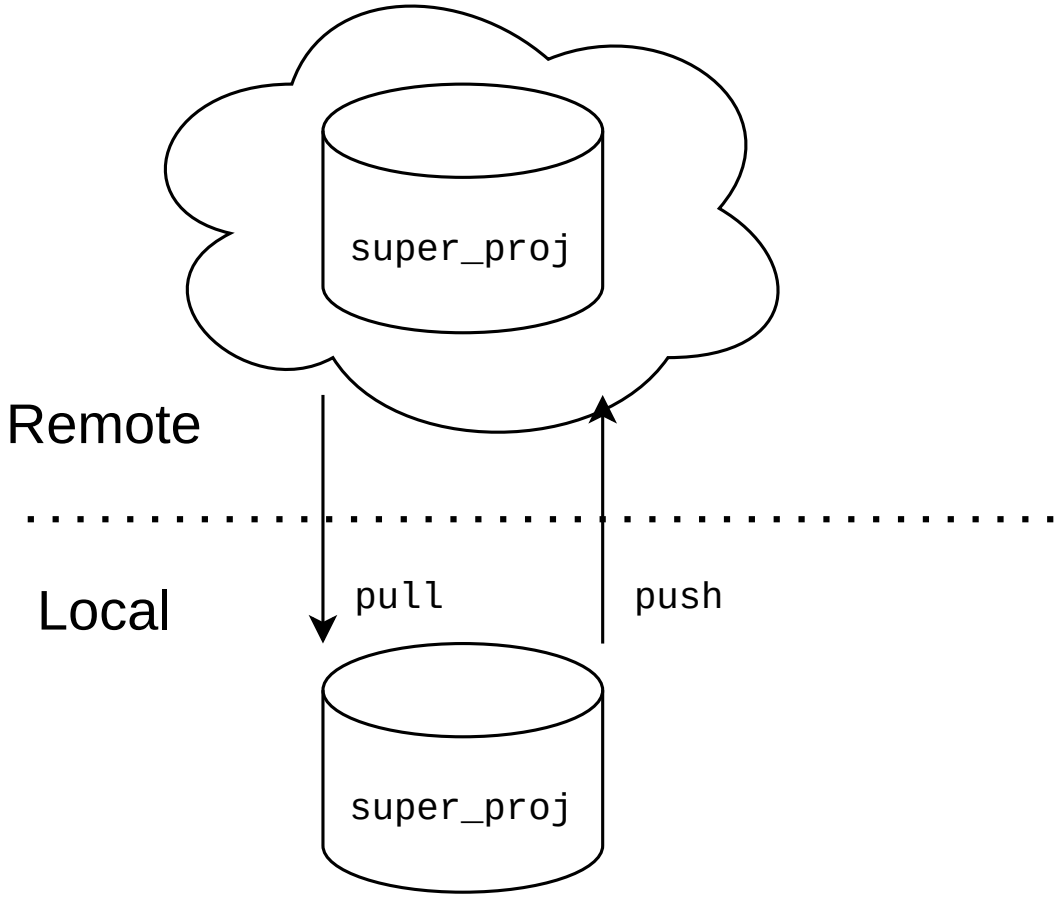


Single user, multiple branches

- `git checkout -b dev` (shortcut for `git branch dev && git checkout dev`)
- `git checkout main`
- `git merge dev`

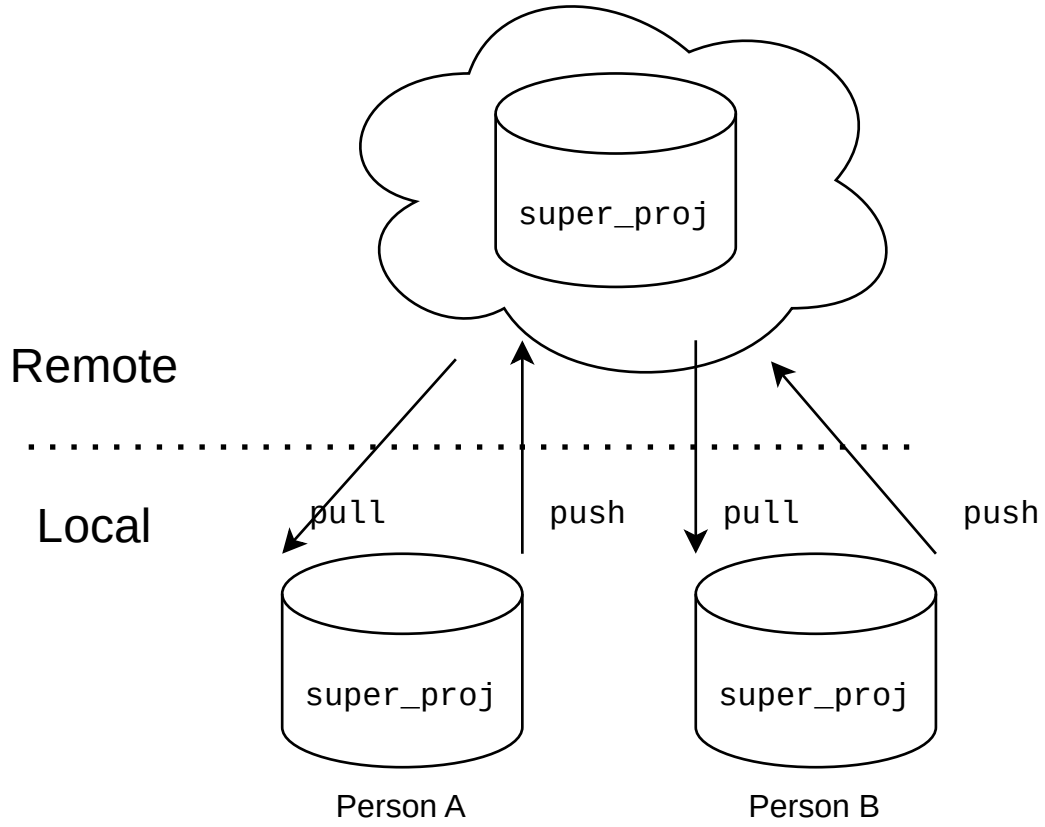


Adding a remote



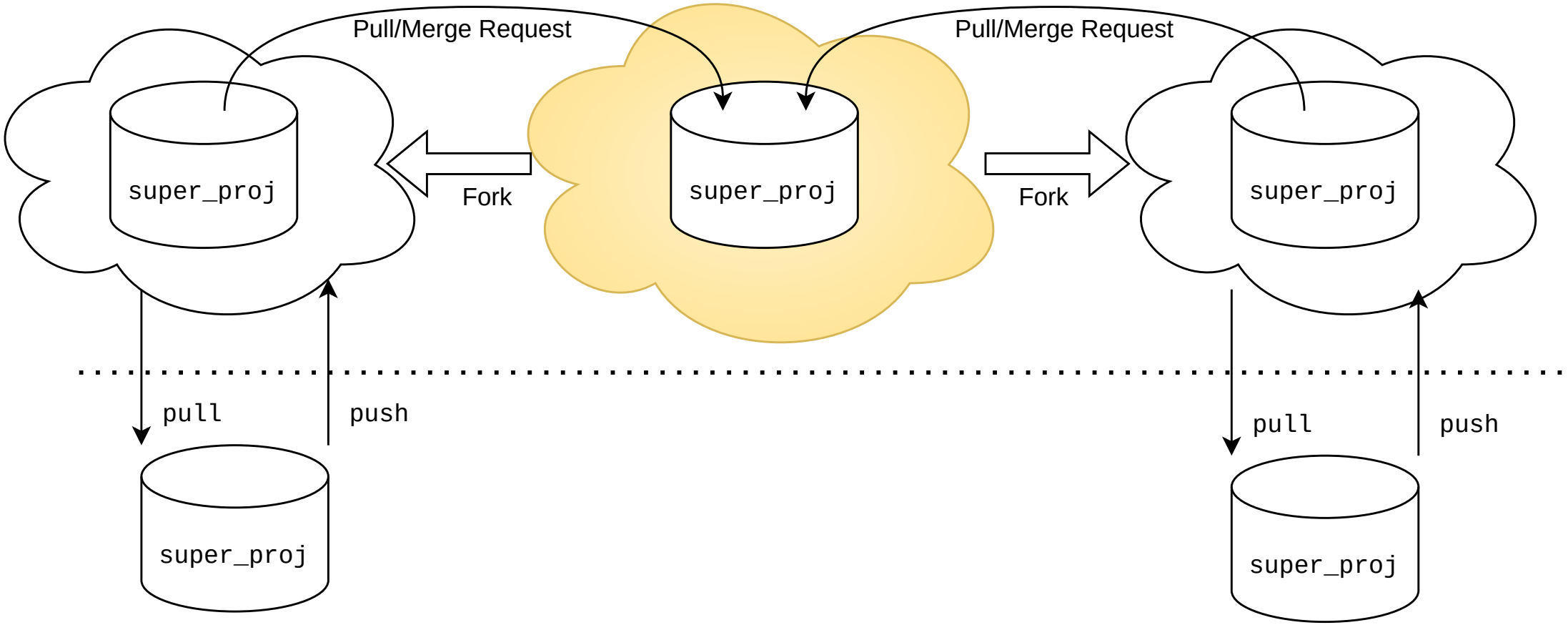
- `git remote add <remote_name> <url>`
- If you cloned from URL at the start, the remote `origin` should exist
- `git remote -v` shows all defined remotes and their roles

Multiple users, single remote, single branch



- Same commands, but now risk of **merge conflicts**
- Might need `git rebase`
- `git pull` is actually short for `git fetch` then `git merge` *or* `git rebase`, depending on configuration

Multiple users, multiple forks, single branch



Multiple users, multiple forks, multiple branches

- The most common setup in FOSS development
- One "canonical" repo, each developer has a fork
- Pull/merge requests (PRs) are **branch-specific**
  - Updates to a branch with an open PR will be added to the PR
  - Typical workflow: create a new branch for every new feature/bug fix

> *This is where I had you jump into the deep end last Thursday*

Exploring communities — Wednesday

- With the projects you posted on the last homework discussion:
  - What are some common documents?
  - How do you evaluate project activity?
  - How is communication handled?

> ***Homework**: Sign up to write class notes for a topic*